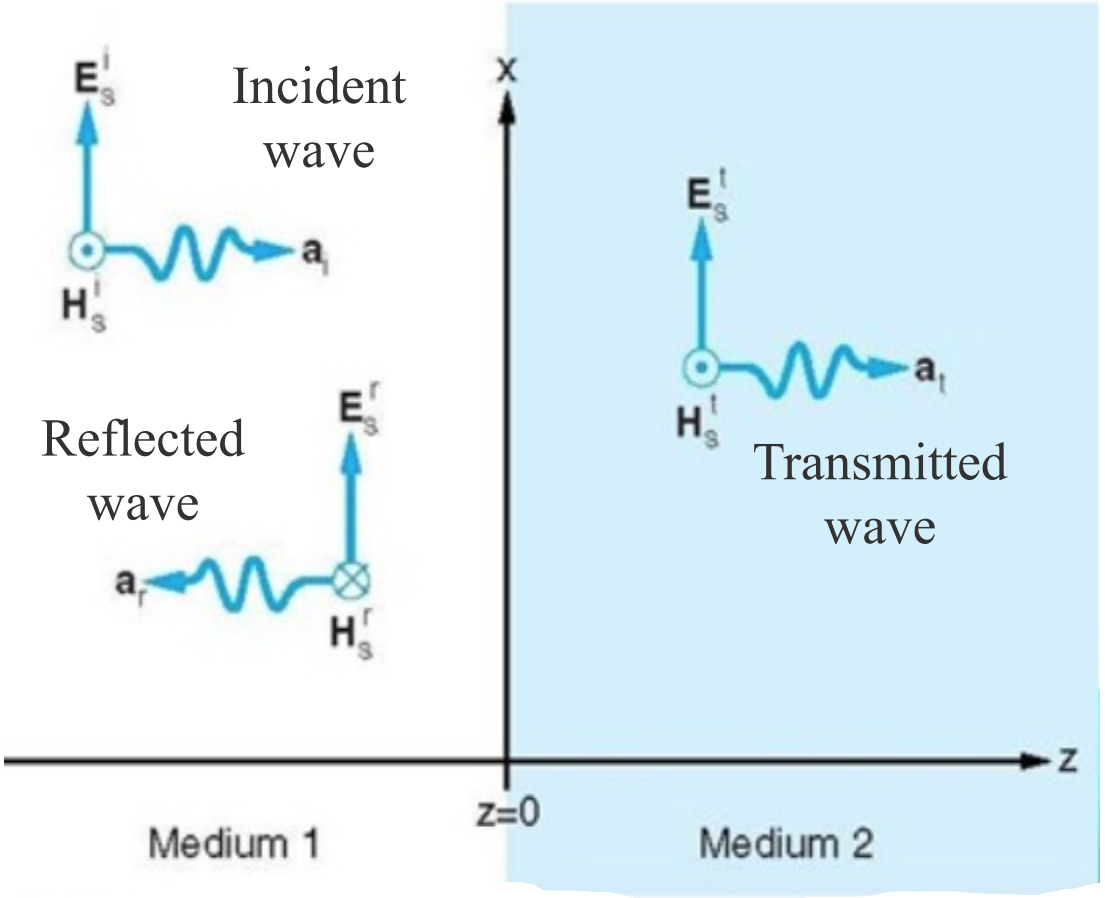
EM Waves Reflection and Transmission

The following subjects will be covered

- Single interface, Normal Incidence.

Single Interface: Normal Incidence



$\varepsilon_1, \mu_1, \sigma_1, \eta_1, \gamma_1$ $\varepsilon_2, \mu_2, \sigma_2, \eta_2, \gamma_2$

Single Interface: Normal Incidence

Incident Wave

$$\overrightarrow{E_i}(z) = E_o\, e^{-j\beta_1 z}\, \vec{a}_x$$

$$\overrightarrow{H_i}(z) = \frac{E_o}{\eta_1} e^{-j\beta_1 z}\, \vec{a}_y$$

Reflected Wave

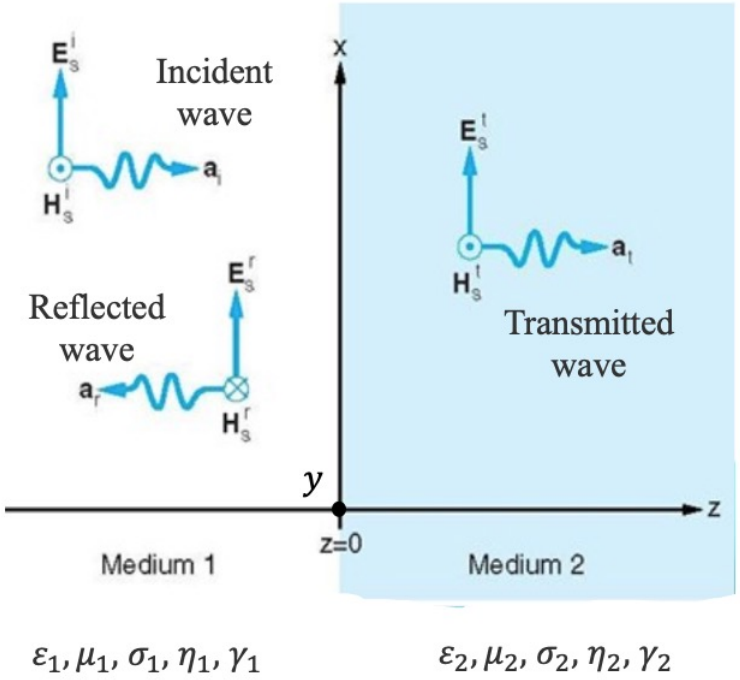
$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+j\beta_1 z}\, \vec{a}_x$$

$$\overrightarrow{H_r}(z) = \frac{\Gamma E_o}{\eta_1}\, e^{+j\beta_1 z}\ \ (-\vec{a}_y)$$

Transmitted Wave

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-j\beta_2 z}\ \ \vec{a}_x$$

$$\overrightarrow{H_t}(z) = \frac{\tau\, E_o}{\eta_2}\, e^{-j\beta_2 z}\ \ \overrightarrow{a_y}$$



$\varepsilon_1, \mu_1, \sigma_1, \eta_1, \gamma_1$ $\varepsilon_2, \mu_2, \sigma_2, \eta_2, \gamma_2$

Apply boundary conditions at interface $z = 0$

$$E_{tangential\ (1)} = E_{tangential\ (2)}$$

$$H_{tangential\ (1)} = H_{tangential\ (2)}$$

Single Interface: Normal Incidence

Apply boundary conditions

$$E_{1_tan} = E_o(1+\Gamma)$$

$$E_{2_tan} = E_o\tau$$

$$\Gamma = 1 - \tau$$

$$H_{1_tan} = \frac{E_o}{\eta_1}(1-\Gamma)$$

$$H_{2_tan} = \frac{\tau E_o}{\eta_2}$$

$$\tau = \frac{2\eta_2}{\eta_1 + \eta_2}$$

Reflection coefficient

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

Transmission coefficient

Power coefficients

$R = |\Gamma|^2$ Reflectance

$T = \frac{\eta_1}{\eta_2}|\tau|^2$ Transmittance

$R + T = 1$ Power Conservation

1) Γ and τ are in general complex.

2) These are field reflection/transmission coefficients.

3) $0 \leq \Gamma \leq 1$

Single Interface: Normal Incidence

Reflection coefficient

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

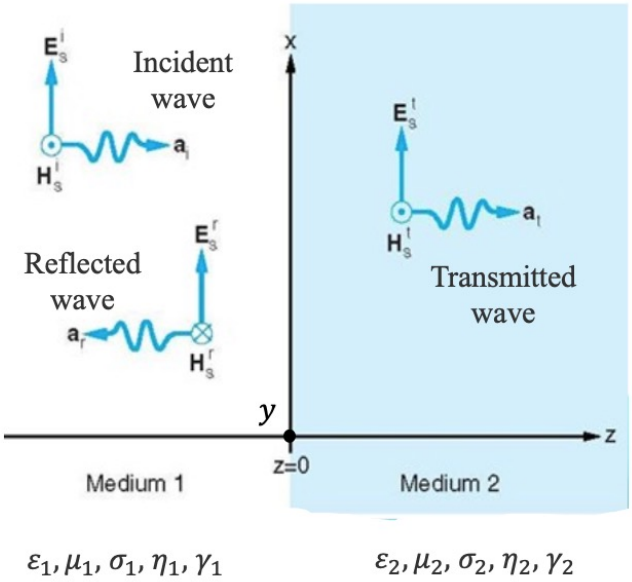
Transmission coefficient

$$\tau = \frac{2\eta_2}{\eta_1 + \eta_2}$$

$$\Gamma = \tau - 1$$

$$0 \le |\Gamma| \le 1$$

All apply at the interface



Power coefficients

$R = |\Gamma|^2$ Reflectance

$T = \frac{\eta_1}{\eta_2}|\tau|^2$ Transmittance

$R + T = 1$ Power Conservation

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2\eta_1}\,\vec{a}_k$$

$$\vec{S}^r_{ave} = R\frac{|E_o|^2}{2\eta_1}\,(-\vec{a}_k)$$

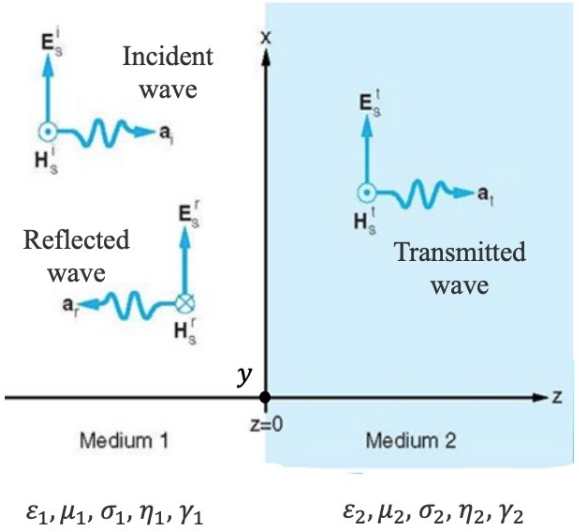
$$\vec{S}^t_{ave} = T\frac{|E_o|^2}{2\eta_1}\,\vec{a}_k$$

Remarks

1) Medium 1 perfect dielectric (η_1=real) and medium 2 perfect conductor (η_2=0)

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -1$$ Total reflection

$$\tau = \frac{2\eta_2}{\eta_1 + \eta_2} = 0$$ Zero transmission



$\varepsilon_1, \mu_1, \sigma_1, \eta_1, \gamma_1$ $\varepsilon_2, \mu_2, \sigma_2, \eta_2, \gamma_2$

2) Medium 1 is a denser dielectric than medium 2

$\varepsilon_{r1} > \varepsilon_{r2}$ $\eta_2 > \eta_1$ $\Gamma > 0$

3) Medium 1 is a lighter dielectric than medium 2

$\varepsilon_{r1} < \varepsilon_{r2}$ $\eta_1 > \eta_2$ $\Gamma < 0$

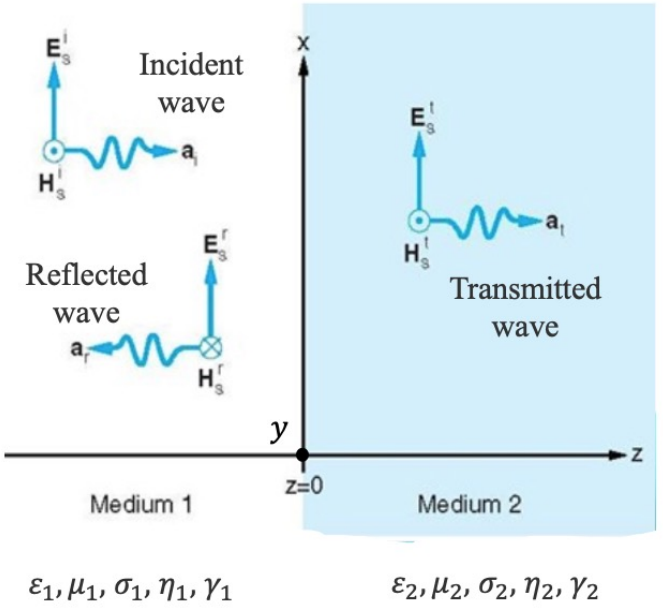
Lossy Media

Electric Fields

$$\overrightarrow{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-j\beta_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+j\beta_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-j\beta_2 z}\, \vec{a}_x$$



Reflection coefficient

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

All apply at the interface

Transmission coefficient

$$\tau = \frac{2\eta_2}{\eta_1 + \eta_2}$$

(η_1 and η_2 complex quantities)

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right) \vec{a}_z$$

$$\vec{S}^r_{ave} = |\Gamma|^2 \frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right) (-\vec{a}_z)$$

$$\vec{S}^t_{ave} = |\tau|^2 \frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{\eta_1}{\eta_2}\right) \vec{a}_z$$

Single Interface: Normal Incidence

Example: Glass Reflection

Glass dielectric constant, $\varepsilon_r \approx 2.3$

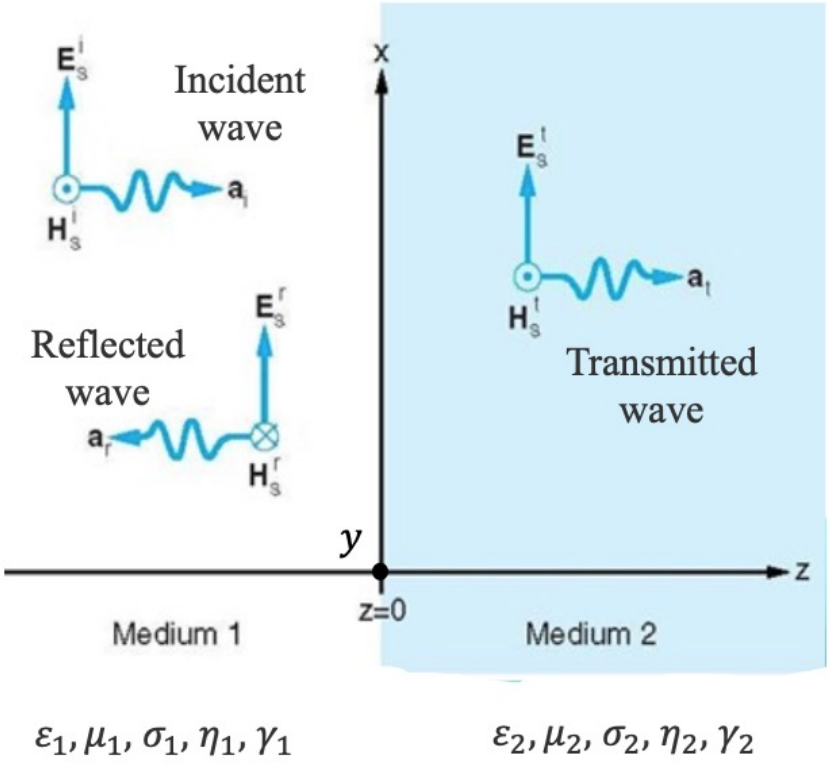
$$\eta = \eta_o \sqrt{\frac{\mu_r}{\varepsilon_r}} = \eta_o \sqrt{\frac{1}{2.3}} = \frac{\eta_o}{1.52}$$

$$\Gamma = \frac{\frac{\eta_o}{1.52} - \eta_o}{\frac{\eta_o}{1.52} + \eta_o} = -0.2065$$

$$R = 0.2065^2 = 0.0426$$

$$T = 1 - 0.0426 = 0.957$$

96% of light power goes through!



$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

Single Interface: Normal Incidence

Example: Seawater Reflection

Seawater dielectric constant, at 2.45 GHz and 30 °C is $\varepsilon_r = 75.9 - j8.6$

$$\eta_2 = \frac{\eta_o}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{75.9 - j8.6}} = 43.1 \angle 3.2^\circ \ \Omega$$

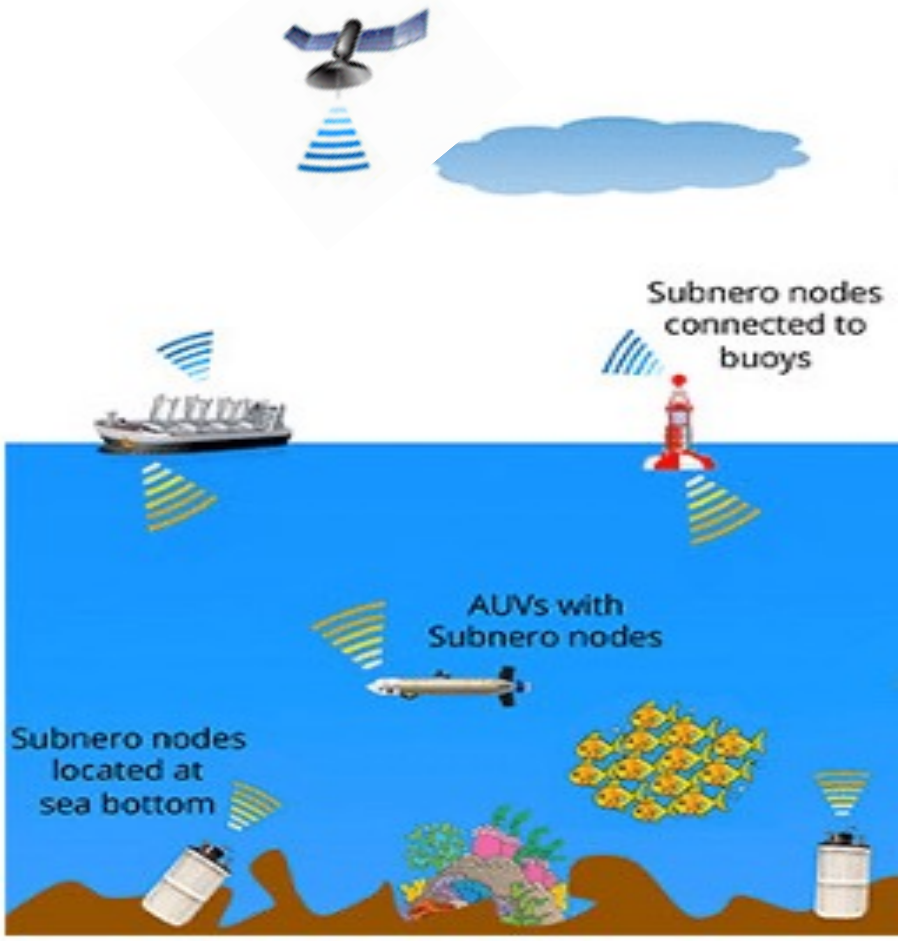
$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\Gamma = \frac{43.1 \angle 3.2^\circ - 377}{43.1 \angle 3.2^\circ + 377}$$

$$\Gamma = \frac{-333.97 + j2.4}{420 + j2.4} = 0.8 \angle -0.74^\circ$$

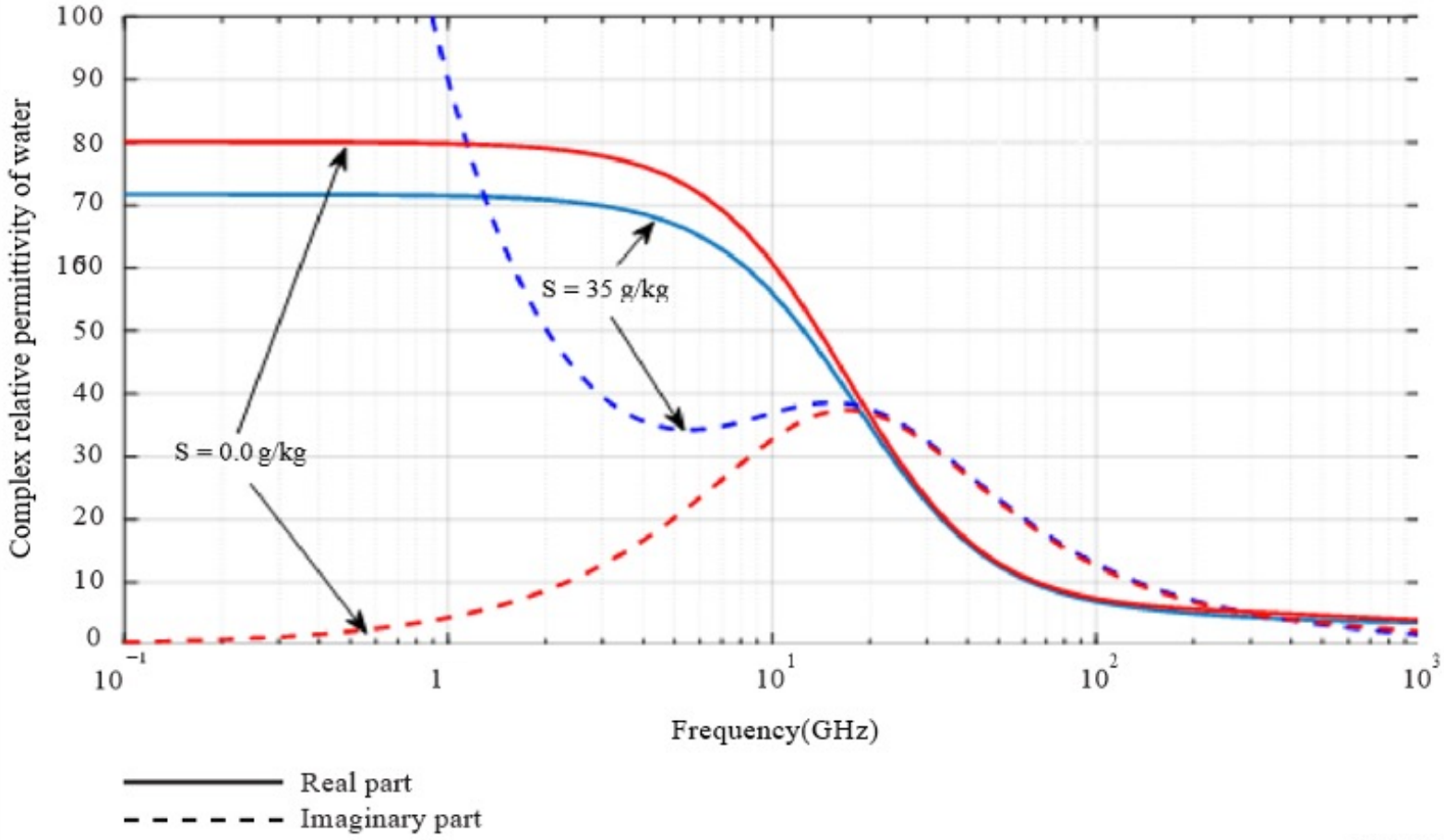
$$R = 0.8^2 = 64\%$$

$$T = 36\%$$



Single Interface: Normal Incidence

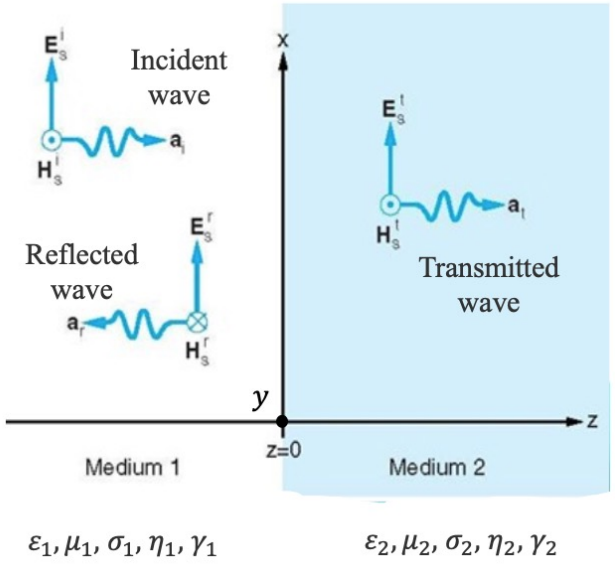
Example: Seawater Reflection



Lossy Media

Example: Lossy dielectric reflection.

A plane wave in air at 1 GHz impinges normally on the surface of a lossy non-magnetic dielectric. The impedance of this dielectric at this frequency is $\eta = 214 + j82.3\ \Omega$. Find the reflection and transmission coefficients. Also, find the reflected and transmitted time-average power density vectors one meter away from the interface. Take $E_o = \sqrt{2}$ kV/m.



For medium 2,

$$\frac{\sigma}{\omega\varepsilon} = \tan 2\theta_\eta = \tan 42^\circ = 0.9$$

Loss tangent ≈ 1

Loss cannot be neglected!

$$\varepsilon_r = 2.0$$

$$\alpha_2 = 12.3 \text{ Np/m}$$

$$\Gamma = -0.2505 + j0.1739$$

$$\tau = 0.7495 + j0.1739$$

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2} Re\left(\frac{1}{\eta_1}\right) e^{-2\alpha_1 z}\, \vec{a}_z = 2.65\, \vec{a}_z \text{ kW/m}^2$$ Constant in air.

$$\vec{S}^t_{ave} = |\Gamma|^2 \frac{|E_o|^2}{2} Re\left(\frac{1}{\eta_1}\right) e^{+2\alpha_2 z}\, (-\vec{a}_z) = 0.246\, (-\vec{a}_z) \text{ kW/m}^2$$

$$\vec{S}^t_{ave} = |\tau|^2 \frac{|E_o|^2}{2} Re\left(\frac{\eta_1}{\eta_2}\right) e^{-2\alpha_2 z}\, \vec{a}_z = 0.5\, \vec{a}_z \text{ nW/m}^2$$

Q & A

Send me an email if you have any questions.